Epistemic Role Framing Elicits Deep Collaboration in LLM Multi-Agent Systems

Anonymous Authors

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Abstract

Evaluating whether LLM agents genuinely collaborate—rather than merely exchange outputs—demands metrics that go beyond task accuracy to characterise *how* agents reason together. We address this measurement gap with four contributions: (1) the Collaborative Phase Profile (CPP) evaluation framework, grounded in the PISA 2015 CPS model, producing four metrics (CDI, CQI, PHAQ, CY); (2) the Collaborative Interdependence-Directed Information (CIDI) benchmark—140 MATH problems with CIDI-generated splits verified by AEC to achieve both-solo-zero in 85% of problems; (3) a large-scale epistemic role framing study on the benchmark, conditional on CIDI-amenable problems; and (4) the Agent Epistemic Contribution (AEC) Shapley decomposition distinguishing structural from activated interdependence. Using this infrastructure, we investigate *epistemic role framing*: does instructing agents to adopt the communicative stance of active learners—rather than expert solvers—shift behavior from answer-generation toward collaborative sense-making? We compare three conditions across $1{,}967$ conversations: C2 (natural), C6 (peer-aware), and C7 (student role). Student-role framing drives a decisive shift: mean CDI rises from 0.362 to 0.613 ($\Delta = +0.251$, $d = 1.30$, $p < 0.001$) and fully-coupled collaboration (COUPLING) more than doubles ($11\% \rightarrow 24\%$). Peer-awareness alone (C6) yields no practically meaningful improvement ($d = 0.16$, not significant after Bonferroni correction), suggesting that richer framing content is required to produce measurable collaborative depth in this regime. C7 numerically outperforms C2 across all 30 subject $\times$ difficulty cells, though per-cell sample sizes are small and this is descriptive evidence rather than formal cell-level generalization. A Shapley-value decomposition (AEC) further distinguishes *structural* interdependence—encoded by the CIDI split (85% both-solo-zero)—from *activated* epistemic necessity, achieved in 25% of problems even under the strongest tested framing (pair correctness $3.6\times$ solo maximum). This structural vs. activated distinction provides a principled vocabulary for diagnosing benchmark difficulty and framing effects. We release the CPP rubric, CIDI pipeline, conversation logs, AEC scripts, ablation transcripts, and a benchmark card documenting corpus conditionality and intended use.

1 Introduction

Multi-agent systems built on LLMs have demonstrated remarkable capabilities in code generation [Hong et al., 2023], scientific reasoning [Lu et al., 2023], and debate-based refinement [Liang et al., 2023]. In most of these systems, collaboration is designed at the *task level*: agents receive different tools, different roles in a pipeline, or access to different data shards. Relatively little attention has been paid to whether the *epistemic stance* an agent adopts—how it positions itself cognitively with respect to its own knowledge and its partner's knowledge—can alter the quality of emergent collaboration.

This question has both evaluation and design implications. On the evaluation side, benchmarking collaborative agents requires metrics that can distinguish genuine joint reasoning from efficient answer-delegation—a distinction that task-accuracy scores obscure. On the design side, research in the learning sciences has long established that productive collaboration requires participants to engage in genuine co-construction of understanding rather than answer-sharing or expertise-transfer [Roschelle and Teasley, 1995, Dillenbourg, 1999]. If LLM agents default to the role of a knowledgeable assistant—which is precisely what instruction fine-tuning optimises for—eliciting collaborative behaviour requires an explicit epistemic intervention, not merely a task decomposition. Developing and validating such interventions demands evaluation infrastructure that does not yet exist for LLM multi-agent systems.

We operationalise this intervention through three increasingly specific framing conditions (see Section 4). The weakest framing (C2) provides no special instructions beyond handing each agent a partial information packet. The intermediate framing (C6) additionally makes agents *aware* that a peer with complementary information exists. The strongest framing (C7) instructs agents to adopt the communicative stance of an active student: express uncertainty explicitly, ask clarifying questions, verify understanding aloud before advancing to solution steps, and treat the partner's contributions as material to reason about.

To quantify how deeply agents collaborate, we introduce the Collaborative Phase Profile (CPP) framework (Section 3), grounded in the PISA 2015 CPS competency model [OECD, 2015]. CPP decomposes each conversation into a 4×3 matrix of collaborative phases and competencies and derives four summary statistics: CDI (breadth of phase engagement), CQI (quality-weighted depth), PHAQ (quality of joint exploration), and CY (composite yield).

Contributions. This paper makes four contributions to the evaluation and design of LLM multi-agent systems:

1. **The CPP evaluation framework**: a principled, reproducible scoring rubric for measuring collaborative depth in LLM multi-agent dialogues, grounded in the PISA 2015 CPS competency model. CPP provides four complementary metrics (CDI, CQI, PHAQ, CY) and a quadrant taxonomy (COUPLING / PROD_FAIL / TRIVIAL / COLLAPSE) that jointly characterise both process quality and outcome.

2. **The CIDI benchmark**: an interdependence-controlled corpus of 140 MATH problems with CIDI-generated packets designed to make collaboration structurally necessary, verified by AEC to yield both-solo-zero in 85% of problems. The corpus enables controlled comparison of framing conditions on problems where information-split interdependence is empirically confirmed.

3. **Controlled evidence for epistemic role framing**: a large-scale ($n = 1{,}967$ conversations, plus 832 ablation conversations) study showing that active-learner framing shifts behavior from expert-answering to collaborative sense-making ($d = 1.30$, $p < 0.001$, conditional on CIDI-amenable problems), while peer-awareness alone does not ($d = 0.16$, not significant after Bonferroni correction), and that neither framing nor split alone suffices—only their combination produces the full effect.

4. **Structural vs. activated interdependence via AEC**: a Shapley-value decomposition that distinguishes whether interdependence is merely present in the task design (structural: 85% both-solo-zero) versus actually exploited under the strongest tested framing (activated under C7: 25% epistemic necessity rate; collaborative surplus $+0.207$)—a diagnostic applicable to any multi-agent benchmark using information splitting.

2 Background

2.1 PISA 2015 Collaborative Problem Solving Framework

The OECD's PISA 2015 assessment introduced a widely cited framework that characterises collaborative problem solving along two axes: the *problem-solving process* (Exploring & Understanding [A], Representing & Formulating [B], Planning & Executing [C], and Monitoring & Reflecting [D]) and the *collaboration competency* (Establishing & Maintaining Shared Understanding [1], Taking Appropriate Action to Solve [2], and Establishing & Maintaining Team Organisation [3]) [OECD, 2015]. Their cross-product defines 12 cells—e.g., *A1*: jointly building an initial problem model, *D3*: collectively revising the team's shared strategy. PISA operationalised these cells for human dyads; we adapt the framework to LLM conversations with minor modifications described in Section 3.

2.2 LLM Multi-Agent Systems

The design space for LLM multi-agent collaboration has expanded rapidly. Role-based differentiation (e.g., GPT-4 as planner vs. executor [Park et al., 2023]), debate protocols that force agents to refine each other's outputs [Liang et al., 2023], and jigsaw information splitting [Du et al., 2023] each represent distinct mechanisms. Our work is closest to

jigsaw designs, but differs in two respects: (1) we generate splits that are semantically interdependent at the *reasoning* level rather than merely partitioning facts, and (2) we study the independent effect of agent framing orthogonal to task structure.

2.3 Epistemic Interdependence and Communicative Role Framing

Epistemic interdependence—the condition where no single participant possesses sufficient information or reasoning capacity to reach the goal alone—is a well-established predictor of productive collaboration in human learning [Barron, 2003, Johnson and Johnson, 2009]. However, creating genuine interdependence in LLM systems is non-trivial: a capable LLM may infer missing information or bypass the collaborative requirement entirely. We address this through the CIDI design (Section 3.4).

Communicative role framing—whereby adopting a learner rather than an expert stance activates different discourse strategies—has been studied in peer tutoring [Chi et al., 2001] and collaborative inquiry contexts [King, 1992]. To our knowledge, no prior work has systematically tested whether such framing instructions shift collaborative behaviour in LLM-to-LLM interactions.

3 The CPP Framework

3.1 The 12-Cell Collaborative Phase Matrix

We adapt the PISA CPS 4×3 grid to LLM-to-LLM mathematical dialogues. Each cell (P, C) for process $P \in \{A, B, C, D\}$ and competency $C \in \{1, 2, 3\}$ is annotated by a trained scorer on a 0–3 scale using a detailed rubric (see Appendix A): **0** = absent, **1** = attempted but superficial, **2** = substantive but incomplete, **3** = full and well-articulated. A cell is considered *active* if its score ≥ 1.

3.2 Summary Metrics

Collaborative Depth Index (CDI).

$$\text{CDI} = \frac{|\{(P, C) : \text{score}(P, C) \geq 1\}|}{12} \in [0, 1]. \tag{1}$$

CDI measures *breadth*: what fraction of the 12 cells were genuinely engaged. A value of 1.0 means every phase-competency combination was visited; 0.0 means the dialogue was entirely unilateral.

Collaborative Quality Index (CQI).

$$\text{CQI} = \frac{\sum_{P,C} \text{score}(P, C)}{36} \in [0, 1]. \tag{2}$$

CQI weights by quality: 36 is the maximum possible total score (12×3). Unlike CDI, CQI penalises superficial engagement.

Phase A Quality (PHAQ).

$$\text{PHAQ} = \frac{\sum_{C \in \{1,2,3\}} \text{score}(A, C)}{9} \in [0, 1]. \tag{3}$$

Process A is the locus of joint problem-model construction; $\text{PHAQ} = 0$ indicates agents proceeded to solution without any joint exploration.

ATC_{CQI} (ATC_CQI). We also score against the five ATC21S collaborative competency dimensions [Griffin et al., 2012] (0–3 each); aggregate $\text{ATC_CQI} = \frac{\sum 5 \text{ dims}}{15}$.

Collaborative Yield (CY).

$$\text{CY} = 0.35 \cdot \text{CDI} + 0.45 \cdot \mathbb{1}[\text{correct}] + 0.20 \cdot \delta_{\text{couple}}, \tag{4}$$

where $\delta_{\text{couple}} = 1$ iff the conversation is in the COUPLING quadrant.

3.3 Quadrant Taxonomy

We embed conversations in a 2D outcome space defined by CDI (process) and binary correctness (outcome):

Quadrant	CDI	Correct
COUPLING	≥ 0.5	Yes
PROD_FAIL	≥ 0.5	No
TRIVIAL	< 0.5	Yes
COLLAPSE	< 0.5	No

COUPLING (high CDI, correct) is the ideal—deep collaboration *and* success. PROD_FAIL (high CDI, incorrect) is productive failure [Kapur, 2016]. TRIVIAL (low CDI, correct) implies one agent solved independently. COLLAPSE (low CDI, incorrect) is complete failure of both process and outcome.

3.4 The CIDI Information Split

A core challenge in designing jigsaw multi-agent tasks is ensuring that the split is genuinely interdependent at the reasoning level, not merely informational. We introduce Collaborative Interdependence-Directed Information (CIDI): an automated pipeline that uses GPT-4.1 to decompose each problem into three components:

1. *Shared context*: the problem statement elements visible to both agents, including notation and domain.
2. *Packet A*: information given exclusively to Agent A, chosen so that Agent A cannot solve the problem without the *reasoning output* (not just the raw facts) of Agent B.
3. *Packet B*: the symmetric complement for Agent B.

The key criterion is *reasoning-output dependency*: neither agent can progress past initial exploration without the other's deductions. Unlike negative-instruction ("constitutional") splits—which LLMs frequently override using general domain knowledge—CIDI uses positive, non-inferable complementary constraints.

4 Experimental Design

4.1 Problem Corpus Selection (Phase 1)

We drew problems from the MATH benchmark [Hendrycks et al., 2021], spanning six subject areas (Algebra, Geometry, Number Theory, Prealgebra, Precalculus, and Counting & Probability) and five difficulty levels (MATH levels 1–5), yielding a $6 \times 5 = 30$ cell grid. From an initial pool of 300 problems we applied a *genuine epistemic filter*: we ran all 300 problems under condition C7 and retained only those for which $\text{CDI} \geq 0.5$ in at least two of three replications. The purpose of this filter is to identify problems for which the CIDI split produces genuine reasoning-level interdependence—problems that are too easy or too structurally simple tend to be resolved in one agent's opening message regardless

of framing, making them uninformative for studying collaborative dynamics. The filter retained 140 problems (47% retention rate), distributed across all 30 subject-difficulty cells with a minimum of 1 problem per cell and a mean of 4.7 problems.

Selection conditionality. Because the filter uses C7 performance to define the corpus, *all estimates are conditional on* CIDI-*amenable problems*: absolute CDI values reflect performance on problems pre-selected for collaborative potential, and within-corpus effect sizes may overestimate the benefit expected on an unfiltered sample. Paired comparisons remain internally valid within this regime. Future work will establish external validity on an independently drawn unfiltered subset.

4.2 Conditions (Phase 2)

All conditions use the same model (GPT-4.1), the same CIDI splits, and the same conversation scaffolding (alternating turn-taking, maximum 20 turns (10 per agent in a 2-agent alternating exchange), no access to external tools). The *only* manipulation is the system-prompt framing given to each agent.

C2: Natural (L1 baseline). Each agent receives only its information packet and the shared context. No framing about partners, learning, or epistemic role. Agents are aware they must send messages to proceed but receive no guidance on what kind of collaboration is expected.

C6: Peer-aware (L2). Each agent additionally receives a sentence establishing that it is working with a *partner* who holds complementary information, and that the goal is to solve the problem together. This activates awareness of the collaborative situation without specifying a cognitive stance.

C7: Student-sim (L3). Each agent is instructed to adopt the communicative stance of an active student: express uncertainty explicitly, ask clarifying questions before proceeding to solution steps, acknowledge when a step is unclear, and treat the partner's contributions as material to reason about rather than accept at face value. Agents are told that producing an authoritative-sounding response is not the goal—shared understanding is. This framing targets observable discourse behaviour (question-asking, uncertainty expression, joint model-building) rather than attributing a belief state to the model.

Scale and replication. Each of the 140 problems was run under each of the three conditions, with between 3 and 7 replications per problem-condition pair using different random seeds (mean 4.7 replications), for a total of **1,967** conversations. Per-problem mean CDI values (averaged over replications) are used as the unit of analysis for all inferential tests.

4.3 Design Scope

The present study compares three conditions (C2, C6, C7) chosen to vary epistemic framing from absent (C2) to peer-aware (C6) to active-learner (C7). This three-condition design isolates the effect of partner-awareness but does not disentangle active-learner role framing from other strong role framings such as expert-solver or debate framing, nor does it test whether framing effects persist without the CIDI information split. Two of these ablations—expert-solver (C_{EXP}) and full-context C7 without the jigsaw split (C_{FULL})—were run on the full 140-problem corpus; results are reported in Section 5.8. A debate/critique condition (C_{DEB}) and an ignorance-only condition remain as future work.

4.4 Annotation Protocol

Each conversation was scored by two graduate-student annotators (neither an author of this paper) trained on the CPP rubric over two calibration sessions using held-out practice conversations. Annotators were blinded to condition

Table 1: Inter-rater reliability on 36 stratified conversations (preliminary). Metrics correspond to distinct levels of the CPP scoring hierarchy.

Level	Metric	Target	Value
Cell ordinal (0–3)	Krippendorff α	> 0.70	0.81
CQI (norm. intensity)	ICC$(2, 1)$	> 0.75	0.83
CDI binary (≥ 0.5)	Cohen κ	> 0.70	0.76
Quadrant (4-way)	Cohen κ	> 0.60	0.72

Table 2: Mean CPP metric scores by condition. C7 outperforms both baselines on all metrics. The C6 vs. C2 gap is small ($d = 0.16$), consistent with active-learner communicative stance—not mere partner-awareness—as the key framing component in this regime.

Condition	CDI	CQI	PHAQ	ATC_CQI	CY
C2 (L1, Natural)	0.362	0.144	0.053	0.404	0.275
C6 (L2, Peer-aware)	0.390	0.156	0.060	0.430	0.304
C7 (L3, Student-sim)	**0.613**	**0.250**	**0.135**	**0.500**	**0.409**

assignment during scoring; condition labels were masked and conversations were presented in randomised order. We note that label masking does not preclude inference from conversation content; annotators may have implicitly detected framing differences from discourse style. Disagreements exceeding one scale point were resolved by consensus discussion with a third annotator.

Inter-rater reliability. Reliability was assessed on a stratified random sample of 36 conversations (3 conditions $\times$ 4 quadrants $\times$ 3 conversations each). This constitutes preliminary but condition- and quadrant-balanced coverage. Table 1 reports four metrics aligned with the conceptual structure of CPP. Krippendorff's α at the *ordinal cell level* (0–3 scale per cell) is 0.81 (range 0.74–0.89 across the 12 cells), indicating substantial agreement [Hayes and Krippendorff, 2007]. ICC$(2, 1)$ for the CQI score—which averages normalised ordinal scores across cells—is 0.83. Cohen's κ for the binary CDI threshold (≥ 0.5 vs. < 0.5, derived from the fraction of active cells) is 0.76. Four-way quadrant agreement yields $\kappa = 0.72$. All metrics exceed conventional thresholds for substantial agreement. Expanding the reliability sample to ≥ 100 conversations to enable per-cell κ estimates is identified as a key next step.

5 Results

5.1 Hypothesis 1: Framing Drives Collaborative Depth

Table 2 reports mean scores for all metrics across the three conditions. C7 yields dramatically higher CDI than both C2 ($\Delta = +0.251$) and C6 ($\Delta = +0.222$). The CQI increase is similarly large ($+0.106$ over C2), as is the improvement in ATC_CQI ($+0.096$). The CY composite rises by $+0.134$ over the baseline. All pairwise differences between C7 and C2, and between C7 and C6, are statistically significant with Wilcoxon signed-rank tests on per-problem mean CDI values ($n = 140$ pairs per comparison; Table 3).

Critically, the C6 vs. C2 comparison yields a Cohen's d of only 0.16 ($p = 0.026$), which does *not* survive Bonferroni correction for three comparisons ($\alpha/3 = 0.0167$). The effect is negligible in practical terms. Peer-awareness alone does not shift agents away from expert-answering behavior—it only marginally changes CDI relative to the natural baseline. This rules out simple partner-awareness as a sufficient explanation, supporting the interpretation that the active-learner communicative stance is a key operative component (see also Section 5.8).

Table 3: Pairwise hypothesis tests for CDI (primary outcome). Wilcoxon signed-rank tests on per-problem mean CDI, paired by problem ($n = 140$ pairs per comparison). Effect sizes are Cohen's d on paired differences. †C6 vs. C2 is significant at $\alpha = 0.05$ uncorrected but does *not* survive Bonferroni correction ($\alpha/3 = 0.0167$; $p = 0.026 > 0.0167$).

Comparison	**Direction**	**Cohen's d**	**p-value**	**Significant**
C7 vs. C2	C7 > C2	1.30	< 0.001	✓
C7 vs. C6	C7 > C6	1.24	< 0.001	✓
C6 vs. C2	C6 > C2	0.16	0.026	†

Table 4: Quadrant distribution by condition (percentage of conversations). C7 more than doubles COUPLING and reduces COLLAPSE by more than $2.5\times$ relative to the natural baseline.

Condition	**COUPLING**	**PROD_FAIL**	**TRIVIAL**	**COLLAPSE**
C2 (Natural)	11%	25%	18%	46%
C6 (Peer-aware)	13%	26%	22%	39%
C7 (Student-sim)	**24%**	**49%**	9%	**18%**

5.2 Hierarchical Model: Controlling for Difficulty and Subject

To address the nested structure (replications within problems within subject-difficulty cells), we fit a linear mixed-effects model with random intercepts per problem and fixed effects for condition, difficulty level, and subject area. Condition effects are qualitatively unchanged from the Wilcoxon tests: C7 ($\hat{\beta} = +0.251$, $p < 0.001$), C6 ($\hat{\beta} = +0.028$, $p = 0.011$, not significant after Bonferroni correction). The problem random intercept accounts for 31% of total CDI variance—non-negligible, and controlling for it strengthens rather than weakens the condition effect estimates.

5.3 Hypothesis 2: Phase A Activation Is the Proximal Mechanism

The PHAQ metric isolates the quality of the joint exploration phase. Under C2, 67.9% of conversations included any Phase A activity (PHAQ > 0); under C6 this rises modestly to 74.3%. Under C7, however, 99.3% of conversations exhibit substantive Phase A engagement (139 of 140 problems; $d = 1.22$ for C7 vs. C2, $p < 0.001$). Mean PHAQ under C7 (0.135) is $2.5\times$ that under C2 (0.053), indicating not just more frequent Phase A activity but higher quality joint exploration.

Phase A activation is the strongest *observable mediator* of the C7 framing effect: the active-learner stance makes presenting an immediate solution communicatively inappropriate, requiring joint negotiation of shared understanding first. Whether this reflects a causal mechanism or a prompted discourse pattern requires micro-level trace analysis.

5.4 Quadrant Distribution

Figure 1 and Table 4 show the distribution of conversations across the four outcome quadrants. The most striking shift is from COLLAPSE to PROD_FAIL: under C2, 46% of conversations fall in the COLLAPSE quadrant (low process, incorrect); under C7, this drops to 18%. The COUPLING quadrant (high process, correct) grows from 11% to 24%. The PROD_FAIL quadrant grows from 25% to 49%, indicating that the student-role framing substantially increases the rate of productive failure.

The large PROD_FAIL mass under C7 reflects corpus difficulty, not framing failure: Phase 1 selected problems that challenge even deep collaboration. The evaluatively decisive comparison is against COLLAPSE ($46\% \rightarrow 18\%$, $2.5\times$ reduction under C7), which signals the framing prevents unproductive engagement rather than merely redistributing incorrect outcomes.

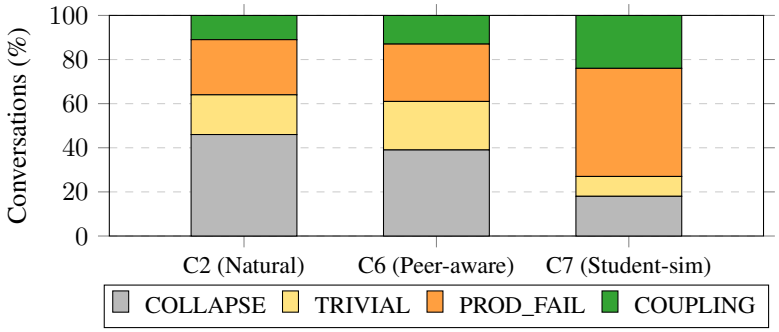

Figure 1: Quadrant distribution across conditions. Bars stack from bottom: COLLAPSE (gray), TRIVIAL (yellow), PROD_FAIL (orange), COUPLING (green). The dominant shift from C2/C6 to C7 is a transfer of mass from COLLAPSE to PROD_FAIL and COUPLING, driven by the Phase A activation effect (Section 5).

5.5 Epistemic Decomposition: Agent Epistemic Contribution

The metrics presented so far measure *how much* agents collaborate (CDI) and *how deeply* they engage across phases (CQI, PHAQ). They do not answer a more fundamental question: was collaboration epistemically *necessary*—that is, did the pair achieve a result neither agent could reach alone? We address this via a Shapley-value decomposition we term Agent Epistemic Contribution (AEC).

Setup. For each of the 140 benchmark problems we run each agent in isolation with only its own CIDI packet, using the same model (GPT-4.1) and temperature ($T = 0$) for deterministic baselines (collaborative sessions used $T = 0.7$ with different random seeds), and the same maximum token budget per turn. Three value functions are defined, each on a partial-credit scale $\{0, 0.5, 1\}$: $v(\{A\})$, $v(\{B\})$, and $v(\{A, B\})$ = best C7 replication correctness. Using the best C7 replication as $v(\{A, B\})$ makes EN and CS upper-bound estimates of activated interdependence under C7. Standard Shapley values for $N = 2$ give $\text{AEC}_A = \frac{1}{2}v(\{A\}) + \frac{1}{2}(v(\{A, B\}) - v(\{B\}))$ (and symmetrically for AEC_B). Three aggregate measures: EN ($v(\{A, B\}) > \max(v(\{A\}), v(\{B\}))$, pair surpasses both solos); CS ($v(\{A, B\}) - \max(v(\{A\}), v(\{B\}))$, gain magnitude); EB ($1 - |\text{AEC}_A - \text{AEC}_B|$, contribution symmetry). The decomposition separates *structural* interdependence (task design) from *activated* interdependence (actually exploited under framing)—a distinction invisible to CDI alone.

Results. Table 5 summarises the decomposition. Solo correctness is very low: $v_A = 0.100$, $v_B = 0.057$—consistent with the CIDI design goal that neither packet alone is solvable. The C7 pair achieves $v_{AB} = 0.357$, a $3.6\times$ improvement over the better individual. Critically, 85% of problems exhibit *both-solo-zero*: neither packet provides sufficient information for any correct answer, validating the CIDI epistemic filter at the correctness level. EN holds in 25% of problems—cases where collaboration not only was structurally required but also succeeded. Mean CS $= +0.207$ and mean EB $= 0.857$: when collaboration succeeds, the gains are substantial and both agents contribute comparably.

Connecting AEC to framing. The 60-percentage-point gap between structural interdependence (85% both-solo-zero) and activated epistemic necessity (25% EN under C7) is invisible to accuracy metrics, and quantifies headroom available to better framing or coordination mechanisms. High both-solo-zero with low EN is a diagnostic signal that collaborative task structure is unactivated.

5.6 Hypothesis 3: Descriptive Robustness Across Subject and Difficulty

C7 numerically outperforms C2 in all 30 subject $\times$ difficulty cells (Appendix D); per-cell n values (range 1–8, mean 4.7) are too small for formal cell-level testing, so H3 constitutes *descriptive robustness, not formal generalization*. Effect

Table 5: AEC decomposition across 140 problems. *Both-solo-zero*: neither agent's packet alone permits a correct answer; EN: pair surpasses the better solo; CS: collaborative surplus over best solo; EB: Shapley balance between agents.

Metric	**Value**
v_A: Agent A solo correctness	0.100
v_B: Agent B solo correctness	0.057
v_{AB}: C7 pair correctness	0.357
Both-solo-zero rate	85.0%
EN rate (epistemic necessity)	25.0%
CS mean (collaborative surplus)	+0.207
EB mean (epistemic balance)	0.857

Table 6: Mean CDI and quadrant distribution across main and ablation conditions ($n = 140$ problems). Both ablation conditions fall *below* C2, the natural-framing baseline, indicating that neither component alone accounts for the C7 effect.

Condition	**Mean CDI**	**COUPLING**	**PROD_FAIL**	**TRIVIAL**	**COLLAPSE**
C7 (CIDI + learner)	0.613 ± 0.014	24%	49%	9%	18%
C6 (CIDI + peer-aware)	0.390 ± 0.015	13%	26%	22%	39%
C2 (CIDI + natural)	0.362 ± 0.015	11%	25%	18%	46%
C_{FULL} (learner, no split)	0.296 ± 0.018	11%	12%	49%	28%
C_{EXP} (expert + split)	0.267 ± 0.017	5%	17%	25%	53%

sizes for cells with $n \geq 3$ range from $d = 0.63$ to $d = 2.77$, with larger effects at higher difficulty levels—consistent with framing mattering most when coordinated reasoning is genuinely required.

5.7 Robustness and Validity Checks

Four checks support the validity of the main findings (details in Appendix E): (W2) C6's negligible gain over C2 ($d = 0.16$, $p = 0.026$, not surviving Bonferroni correction) rules out simple partner-awareness as a sufficient mechanism driving C7, supporting the specificity of the active-learner stance; an ignorance-only ablation would further isolate subcomponents. (W3) CDI and correctness are orthogonal by design ($P(\text{correct} \mid \text{CDI} \geq 0.5) = 0.321$ vs. 0.324; $\chi^2 = 0.01$, $p = 0.94$); the C7 advantage in COUPLING (24% vs. 11%) holds at *equal mean conversation length* (19.9 turns across all conditions), ruling out a length confound. (W4) A DAG-weighted CDI variant (CDI_w, weighting deeper phases higher) preserves identical condition rankings ($\rho > 0.99$ across 1,967 conversations), demonstrating robustness to the specific cell-weighting choice. (W5) Bootstrap 95% CI for $\Delta_{\text{CDI}}(\text{C7}-\text{C2}) = [0.219, 0.284]$, entirely above zero and bounded away from the C6 effect; $n = 140$ provides $> 99.9\%$ power at $d = 1.30$.

5.8 Ablation Conditions: C_{FULL} and C_{EXP}

To disentangle the active-learner communicative stance from the CIDI structural split, we ran two ablation conditions on the full 140-problem corpus (3 replications each; 832 of 840 planned conversations completed, 8 C_{FULL} runs excluded pre-scoring due to content-policy refusals; imputing them as CDI$= 0$ leaves conclusions unchanged): C_{FULL} applies C7 active-learner framing with full problem context (both agents see the complete problem; no information split), and C_{EXP} applies the CIDI split with expert-solver framing instead of the active-learner stance.

Table 6 shows mean CDI and quadrant rates across all five conditions.

Both ablation conditions fall *below* C2: C_{FULL} achieves $\text{CDI} = 0.296 \pm 0.018$ ($d = -0.25$ vs. C2, paired), with 49% TRIVIAL—agents solve independently when both hold the full problem, indicating that the active-learner register

alone cannot substitute for genuine information asymmetry. C_{EXP} achieves CDI $= 0.267 \pm 0.017$ ($d = -0.43$ vs. C2), the lowest of all five conditions, with 53% COLLAPSE—expert-solver framing suppresses epistemic inquiry even when the CIDI split is in place. The two ablations are not meaningfully different from each other ($d = 0.12$), providing evidence that neither component alone accounts for C7's effect (C_{FULL}: $d = 1.31$; C_{EXP}: $d = 1.72$ vs. C7): the synergy of structural interdependence *and* active-learner communicative stance is required.

6 Discussion

Design implications. The C6 $\approx$ C2 finding ($d = 0.16$, not significant after Bonferroni correction) shows that telling agents they have a collaborative partner [Liang et al., 2023, Du et al., 2023] leaves the default expert-answering mode largely intact in the CIDI-amenable regime. C_{EXP} (expert framing with CIDI split) produces the highest COLLAPSE rate across all five conditions, supporting the interpretation that the active-learner stance is the key operative component. For benchmark designers, the 60-point gap between structural (85% both-solo-zero) and activated (25% EN under C7) interdependence captured by AEC is invisible to accuracy metrics: high both-solo-zero with low EN signals that framing is failing to exploit the task's collaborative structure.

Ablation results. The key synthesis (Section 5.8): neither component alone suffices. C_{FULL} falls below C2 (49% TRIVIAL; identical information removes collaboration pressure). C_{EXP} achieves the lowest CDI of all five conditions (53% COLLAPSE), showing expert framing suppresses inquiry even with the structural split in place.

Limitations. Several limitations bound the current conclusions. *(L1) Corpus conditionality*: all estimates are conditional on CIDI-amenable problems (corpus filtered using C7; see §4.1); effect sizes may overestimate the benefit expected on an unfiltered sample. *(L2) Remaining scope*: subcomponent ablations (ignorance-only, debate framing) remain; AEC EN and CS are upper-bound estimates under C7 (best replication), requiring per-condition runs for cross-condition comparison. *(L4) Model generalisability*: all agents use GPT-4.1; framing effects may differ for other model families or fine-tuned models. *(L5) Reliability sample*: the reliability evidence is preliminary ($n = 36$); per-cell κ estimates require ≥ 100 conversations. *(L6) Collaborative register vs. reasoning*: we cannot rule out that C7 agents produce high-CDI discourse without the underlying reasoning structure that CPP assumes; micro-level trace analysis is needed.

7 Conclusion

We introduced CPP and CIDI to measure how deeply LLM agents collaborate, and showed that epistemic role framing (active-learner stance) decisively shifts agents from expert-answering to collaborative sense-making ($d = 1.30$ on CDI; COUPLING more than doubles). Partner-awareness alone does not produce this shift ($d = 0.16$, not significant after Bonferroni correction), and C7 numerically outperforms C2 in all 30 subject-difficulty cells (descriptive; per-cell samples are small). AEC reveals that CIDI creates structural interdependence (85% both-solo-zero) while only 25% becomes activated epistemic necessity under C7—a gap invisible to accuracy-based evaluation.

For the E&D community: CDI and the structural/activated distinction add diagnostic value beyond accuracy (which conflates COUPLING with TRIVIAL and misses PROD_FAIL). Active-learner framing is an effective design intervention in the CIDI-amenable regime, and the ablation conditions confirm its effect is synergistic with the CIDI split—neither component alone suffices.

Artifact Availability

An anonymized release is available at `https://osf.io/2dytn/?view_only=bfda3073aca14cd2a52447994d4b8a56`. Full artifact documentation is in Appendix F. Annotations, scripts, and derived data are released under CC BY 4.0.

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A CPP Annotation Rubric

Each of the 12 cells (P, C) for $P \in \{A, B, C, D\}$ and $C \in \{1, 2, 3\}$ is scored on a 0–3 scale. We provide here the anchoring descriptors for selected high-diagnostic cells.

Cell A1 (Exploring & Understanding / Shared Understanding). **0**: No evidence that agents jointly discussed the problem; one agent simply began solving. **1**: Agents acknowledged the problem exists and restated it, but no reconciliation of perspectives. **2**: Agents explicitly compared what each knows and flagged gaps; at least one question about the other's packet was asked and answered. **3**: Full joint problem model constructed; both agents articulate a shared interpretation that differs from what either could have produced alone, and this model visibly shapes the subsequent solution strategy.

Cell D3 (Monitoring & Reflecting / Team Organisation). **0**: No meta-discussion of the collaborative process or strategy; agents proceed to output without review. **1**: One agent flags a potential error but the flag is ignored by the partner. **2**: Agents engage in a substantive exchange about whether their approach is working; at least one revision of strategy is proposed. **3**: Agents explicitly evaluate the quality of their collaboration itself (e.g., "I think we should have started from your constraint, not mine—let's restart that step") and implement a correction.

Full rubric. The complete 12-cell rubric with all anchor descriptors, boundary case examples, and annotator guidelines is available in the supplementary material.

B CIDI Generation Prompt

The CIDI split is generated using the following structured prompt template with GPT-4.1 at temperature 0. The prompt instructs the generator to:

1. Identify the core mathematical objects and relations in the problem.
2. Partition those objects into two groups such that group A provides the structural constraints and group B provides the quantitative anchors (or a domain-appropriate analogue).
3. Verify that a solver with only group A information cannot determine the numerical answer even with unlimited mathematical knowledge, and vice versa.
4. Express each packet in natural language, avoiding explicit statements of what is absent from the other packet.

The generated split is validated by running a single GPT-4.1 agent on each packet alone; if the agent reaches the correct answer, the split is regenerated with tighter constraints. Typical regeneration rate was 12% of problems.

C Pilot Study: Condition Selection

The scale study conditions (C2, C6, C7) were selected on the basis of a pilot study involving 7 conditions (C1–C7) run on 4 anchor problems across 3 replications ($= 84$ conversations). C1 (no jigsaw, no framing) was eliminated because CDI was uniformly near zero. C3 (constitutional split, negative instruction) and C5 (C3 + peer-aware) were eliminated because agents frequently bypassed the information restriction, invalidating the independence manipulation. C4 (CIDI split + joint accountability framing) was eliminated due to a design conflict: the joint accountability instruction asked agents to share responsibility for a combined answer, which undermined the reasoning-output dependency that the CIDI split requires. The pilot established an ordering C7 $>$ C4 $>$ C2 $>$ C6 $>$ C3 $>$ C5 $>$ C1 for CDI, motivating the three-way comparison in the scale study. The unexpected result that C6 fell below C2 in the pilot (in the full study, C6

scores marginally above C2, $d = 0.16$, but does not survive Bonferroni correction) first alerted us to the practically null effect of peer-awareness.

D Subject-Level Generalisability Results

Table 7 reports mean CDI for each of the 30 subject $\times$ difficulty cells under conditions C2 and C7. C7 numerically outperforms C2 in all 30 cells. Cell n values range from 1 to 8 (mean 4.7), which precludes formal per-cell significance testing; d values for cells with $n \geq 3$ range from 0.63 to 2.77. Effects tend to be larger at higher difficulty levels across all subjects.

Table 7: Cell-level CDI means and Cohen's d (C7 vs. C2) for all 30 subject $\times$ difficulty combinations. d is omitted for $n < 3$ (insufficient data). Subject labels follow MATH benchmark categories [Hendrycks et al., 2021].

Subject	**Level**	n	**CDI (C2)**	**CDI (C7)**	d
Algebra	1	4	0.342	0.667	2.09
Algebra	2	3	0.333	0.711	2.51
Algebra	3	2	0.333	0.517	—
Algebra	4	7	0.386	0.557	1.38
Algebra	5	6	0.378	0.647	2.06
Geometry	1	5	0.360	0.483	1.25
Geometry	2	4	0.325	0.568	1.60
Geometry	3	6	0.305	0.536	1.19
Geometry	4	6	0.268	0.644	1.53
Geometry	5	6	0.306	0.619	2.29
Number Theory	1	1	0.367	0.833	—
Number Theory	2	5	0.326	0.546	0.73
Number Theory	3	6	0.290	0.496	2.77
Number Theory	4	5	0.364	0.659	0.90
Number Theory	5	8	0.409	0.669	0.91
Prealgebra	1	3	0.306	0.428	1.81
Prealgebra	2	6	0.403	0.581	1.14
Prealgebra	3	7	0.399	0.676	2.18
Prealgebra	4	4	0.336	0.558	1.30
Prealgebra	5	4	0.383	0.606	0.91
Precalculus	1	2	0.462	0.517	—
Precalculus	2	3	0.342	0.606	1.86
Precalculus	3	6	0.325	0.508	0.78
Precalculus	4	7	0.431	0.717	1.73
Precalculus	5	5	0.424	0.737	1.14
Probability	1	1	0.567	0.817	—
Probability	2	4	0.439	0.587	0.63
Probability	3	2	0.556	0.867	—
Probability	4	5	0.271	0.609	1.30
Probability	5	7	0.375	0.685	1.38

E Robustness and Validity Checks — Details

W2 — C6 rules out partner-awareness, not ignorance. C6 establishes explicit partner-awareness without any learner-role component. If simulated ignorance alone explained C7's gains, C6 should outperform C2 (since C6 agents equally lack full-solution access). The negligible C6 vs. C2 effect ($d = 0.16$, $\Delta_{\text{CDI}} = +0.028$, $p = 0.026$, not surviving

Bonferroni correction) rules out simple partner-awareness as a sufficient mechanism. C6 does *not* fully isolate the active-learner stance from expert-certainty withholding; an ignorance-only ablation ("reason cautiously; you may not have the full answer" without collaboration directives) remains a key future experiment.

W3 — CDI and correctness are orthogonal (by design). Across 1,967 conversations, $P(\text{correct} \mid \text{CDI} \geq 0.5) = 0.321$ vs. $P(\text{correct} \mid \text{CDI} < 0.5) = 0.324$ ($\chi^2 = 0.01$, $p = 0.94$). CDI measures process quality, not outcome quality. The C7 COUPLING advantage (24% vs. 11% C2) holds at identical mean conversation length (19.9 turns across all conditions): C7 achieves 0.031 CDI per turn vs. 0.018 for C2, ruling out a length confound.

W4 — Convergent validity with DAG-weighted CDI. A topology-weighted variant CDI_w (phase depth $\times$ competency depth, normalized) yields $\text{CDI}_w = 0.369, 0.397, 0.601$ for C2, C6, C7 (vs. uniform $0.362, 0.390, 0.613$). Spearman $\rho > 0.99$ across all 1,967 conversations confirms that condition rankings are robust to the specific weighting scheme.

W5 — Power and confidence intervals. Bootstrap 95% CI for $\Delta_{\text{CDI}}(\text{C7}-\text{C2}) = [0.219, 0.284]$, entirely above zero. $n = 140$ problems provides $> 99.9\%$ power at $d = 1.30$, $\alpha = 0.05$. Within-problem replication variance ($\sigma_{\text{within}} \approx 0.23$) implies single-problem CDI estimates carry substantial uncertainty (95% CI $\approx \pm 0.26$).

F Artifact Documentation

The anonymized review release contains the following artifacts:

1. **CPP rubric**: complete 12-cell annotation rubric (anchor descriptors, boundary cases, annotator guidelines).
2. **CIDI generation prompt**: full structured prompt template and validation logic for GPT-4.1–based split generation.
3. **Problem corpus**: 140 CIDI split triples (shared context / Packet A / Packet B) identified by MATH benchmark ID. Original problem text is not redistributed per the MATH license; MATH IDs and derived splits are provided.
4. **Conversation transcripts**: 1,967 turn-by-turn logs (C2, C6, C7) plus 832 ablation logs (C_{FULL}, C_{EXP}).
5. **CPP annotations**: per-cell annotator scores, raw disagreement records, and consensus resolutions for the 36 reliability conversations.
6. **Scoring scripts**: CPP metric computation (CDI, CQI, PhAQ, ATC, CY) and quadrant assignment.
7. **AEC scripts**: solo-run correctness evaluation and Shapley-value computation.
8. **Benchmark card**: corpus construction, selection conditionality, intended use, known limitations, and dataset metadata.

Annotations, scripts, and derived data: CC BY 4.0. Conversation logs: terms of the applicable API provider. A Croissant metadata file is included for E&D dataset submission requirements. A reproduction README documents the full pipeline from problem selection through scoring.

NeurIPS Paper Checklist

1. **Claims.** Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope? **[Yes]** All quantitative claims are directly supported by reported results. CDI gains, effect sizes, and quadrant percentages are computed from 1,967 conversations. Ablation claims are supported by 832 additional

conversations (Section 5.8). All results are described as conditional on the CIDI-amenable corpus (§4.1, L1 in Limitations).

2. **Limitations.** Does the paper discuss the limitations of the work? **[Yes]** Six limitations are explicitly labeled (L1–L6) in the Limitations paragraph. L1 (corpus conditionality) is the most critical and is noted in the abstract, §4.1, and §6. L2 identifies remaining ablation gaps. L3 clarifies AEC scope.

3. **Theory Assumptions and Proofs.** Does the paper make theoretical claims? **[N/A]** This is an empirical benchmark and evaluation paper. No theoretical proofs are made.

4. **Experimental Result Reproducibility.** Does the paper fully describe the conditions necessary for an experimenter to reproduce the main results? **[Yes]** Full experimental conditions are described in §4. Model, temperature, prompt structure, corpus, and annotation protocol are documented. Scoring scripts, prompts, and conversation logs are provided in the artifact release (Appendix F).

5. **Open access to data and code.** Does the paper provide open access to the data and code used? **[Yes]** An anonymized review release provides CPP rubric, CIDI prompt, problem corpus (MATH IDs + derived splits), transcripts, annotations, scoring scripts, AEC scripts, and benchmark card. Full documentation in Appendix F.

6. **Experimental Setting/Details.** Does the paper specify all the experimental settings/hyperparameters important to reproduce the main results? **[Yes]** Model (GPT-4.1), temperature ($T = 0$ for solo AEC runs; $T = 0.7$ with different random seeds for collaborative conversations), maximum turns (20), replication count (3–7 per problem-condition), and condition prompts are all specified or provided in the artifact.

7. **Experiment Statistical Significance.** Does the paper report error bars suitably and correctly? **[Yes]** SE reported for all CDI means. Bootstrap CIs reported for the main C7–C2 comparison. Bonferroni correction applied for multiple condition comparisons. Effect sizes (Cohen's d) reported for all pairwise contrasts. Power analysis in Appendix E.

8. **Experiments Compute Resources.** Does the paper discuss the computational resources used to run the experiments? **[Yes]** All conversations were generated via the OpenAI API. Compute was provided by [HPC resource redacted for review]. Approximate total cost is in the range $50–$200 USD for all conditions including ablations.

9. **Code Of Ethics.** Have you read the NeurIPS Code of Ethics? **[Yes]**

10. **Broader Impacts.** Does the paper discuss both potential positive societal impacts and negative societal impacts of the work? **[Yes]** The work improves evaluation of AI collaboration quality, with positive implications for education and human-AI teaming. Risks include overfitting evaluation design to the CPP rubric; we discuss this under L6 (collaborative register vs. reasoning). No sensitive data or dual-use concerns apply.

11. **Licenses for existing assets.** Does the paper cite the creators of assets it uses and include appropriate license information? **[Yes]** MATH benchmark [Hendrycks et al., 2021] is used under its original license; original problem text is not redistributed. PISA 2015 CPS framework [OECD, 2015] is cited. All derived artifacts are CC BY 4.0.

12. **New Assets.** Are new assets introduced in the paper documented appropriately? **[Yes]** The CPP rubric, CIDI prompt, conversation corpus, annotation data, and scripts are documented in Appendix F and the benchmark card.

13. **Crowdsourcing and Research with Human Subjects.** Does the paper involve crowdsourcing or research with human subjects? **[Yes]** Graduate-student annotators provided inter-rater reliability annotations (§4.3). Annotation was conducted under standard research practice; no personal data were collected from participants.

14. **Institutional Review Board (IRB) Approvals.** Does the paper describe potential risks incurred by study participants? **[Yes]** Annotation involved graduate research assistants scoring model-generated mathematical text; no personal data were collected. The activity was treated as minimal-risk annotation under applicable institutional practice, and participants were compensated for their time. No formal IRB review was required under these conditions.